Is the LLM judge fair?

arxivkg · evaluation · judge validity

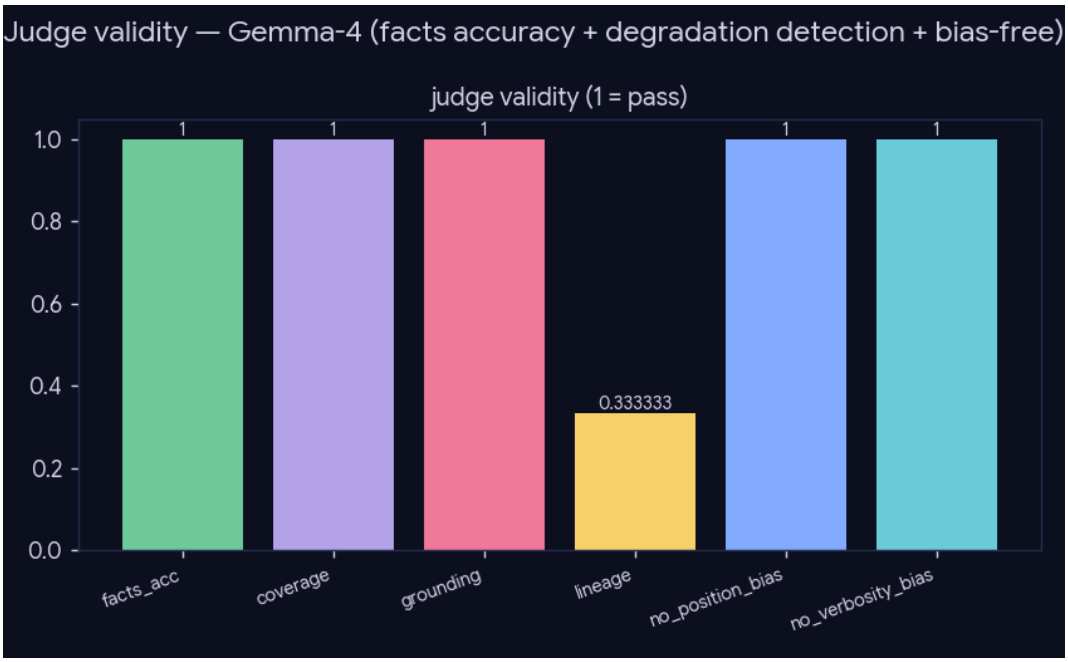
What we did

Quality verdicts rest on an LLM judge (Gemma-4-26B): 4 axes + an order-debiased pairwise. We test it against **ground truth** three ways: (1) **known facts** — 24 claims about famous papers (true / false-topic / false-subtle), accept the true, reject the false; (2) **degradations** — prefer a full review over truncated (coverage), citation-stripped (grounding), chronology-corrupted (lineage) variants; (3) **bias** — identical inputs must tie (position), a content-free padded variant must not win (verbosity). The chronology corruption reverse-maps years so the prose contradicts the dates — an internal inconsistency caught by arithmetic, independent of the judge's stale "now" (today's date is also injected).

Results

Known-fact accuracy: 1.0 (n=24) — it rejects even subtle method-confusions ("DDPM uses a GAN objective" → no; "Transformer relies on RNNs" → no). Coverage detection 3/3, grounding 3/3, lineage 1/3.

degradation	full	tie	variant
identical	0	3	0
truncated-75	3	0	0
truncated-50	3	0	0
truncated-25	3	0	0
citations-stripped	3	0	0
shuffled	3	0	0
chronology-corrupted	1	2	0
verbose-pad	0	3	0



Position bias **0/6 (clean)**; verbosity bias **none**.

Conclusion

The judge is **valid and unbiased on the objective dimensions** — perfect on factual ground truth, catches coverage and grounding degradations at every level, no position or verbosity bias. **Lineage is its weak axis** (catches genuine chronology corruption only sometimes; correctly ignores cosmetic reordering). Rule for every downstream experiment: **trust grounding/coverage and large effects; asterisk lineage and sub-noise-floor margins.** Caveat: small ground-truth set; same-family judging, but a cross-family agreement check found no systematic self-preference.